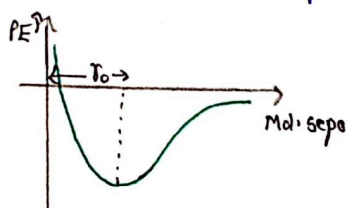


Thermal Expansion:

↳ Caused by ↑ in effective separation due to ↑ in temp.



Temperature Scale ⇒

$$\rightarrow \frac{T - \text{Ice Point}}{(\text{Steam Point}) - (\text{Ice point})} = \text{Const.}$$

$$\rightarrow \frac{K-273}{100} = \frac{^{\circ}C}{100} = \frac{F-32}{180}$$

→ Change in temp. is same in all scales.

Thermal Expansion in Solids ⇒

1) It is photographic i.e. distance b/w each & every point ↑ i.e. π Rm & Pm both ↑ i.e.

2) Linear Expansion: $\Delta L = L \alpha \Delta T$

Area Expansion: $\Delta A = A \beta \Delta T$

Volumetric Expansion: $\Delta V = V \gamma \Delta T$

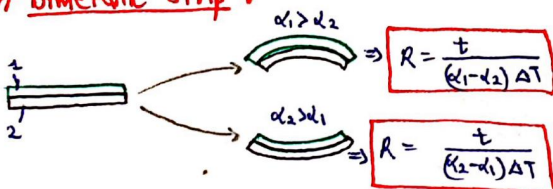
3) For Isotropic material,

$$\alpha : \beta : \gamma \equiv 1 : 2 : 3$$

4) For Anisotropic material,

α_x α_y $\beta = \alpha_x + \alpha_y$

Bimetallic strip:



→ Jiska α zyada, Temp ↑ par increase more
→ Temp ↓ par decrease more

Simple Pendulum:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\frac{\Delta T}{T} = \frac{\alpha \Delta \theta}{2}$$

Thermal Properties of Matter

Thermal strain:

$$\epsilon_{\text{strain}} = \frac{\Delta L}{L \Delta T}$$

$$[\Delta L \ll L] \Rightarrow \epsilon_{\text{strain}} = \frac{\Delta L}{L} = \alpha \Delta T$$

*) Expansion of liquid:

→ For a liq., only 'V' is defined (no α, β)

→ After ΔT

$$V'_L = V_L (1 + \gamma_L \Delta T) \text{ Liquid}$$

$$V'_C = V_C (1 + \gamma_C \Delta T) \text{ container}$$

$$\text{Apparent volume change} = V'_L - V'_C$$

$$\gamma_L = \gamma_C \Rightarrow \Delta V = 0 \Rightarrow \text{Liquid same}$$

$$\gamma_L < \gamma_C \Rightarrow \Delta V < 0 \Rightarrow \text{Liquid drops}$$

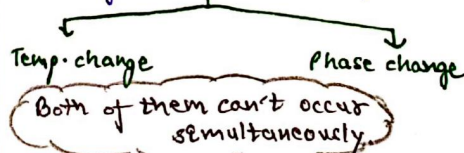
$$\gamma_L > \gamma_C \Rightarrow \Delta V > 0 \Rightarrow \text{Liquid rises}$$

→ On changing temp. by ΔT

$$P_f = P_i (1 - \gamma \Delta T)$$

Calorimetry: [$Q_{\text{gain}} = Q_{\text{loss}}$]

→ Study of changes caused due to heat given to solids & liquids.



1) Specific heat (s) ⇒

Amt of heat req. to ↑ temp. of unit mass by 1 unit temp.

$$s = \frac{Q}{m \Delta T}$$

$$dQ = m \cdot s \cdot dT$$

2) Thermal Capacity (C) ⇒

Amt. of heat req. to ↑ temp. of entire mass by 1 unit temp.

$$C = \frac{Q}{\Delta T} = m \cdot s$$

3) Water Equivalent ⇒

Amt of water that shows same temp change by supplying same amt of heat as of the body.

$$m_w \cdot s_w = m_b \cdot s_b$$

$$1 \text{ cal} = 4.2 \text{ J}$$

$$1 \text{ cal/gm}^{\circ}\text{C} = 4200.57 \text{ J/kg}^{\circ}\text{C}$$

1) Mixing of multiple liquids ⇒

$$T_f = \frac{m_1 s_1 T_1 + m_2 s_2 T_2 + \dots}{m_1 s_1 + m_2 s_2 + \dots}$$

2) In case of emulsion rod/glycer ⇒

$$Q = (P_{\text{out}})(t) = m \cdot s \cdot \Delta T$$

Phase Change:

Latent heat of vaporisation (L_v) ⇒

Amt of heat req. to change 1gm H₂O into 1gm steam at its boiling pt.

$$(L_v)_{\text{water}} = 540 \text{ cal/gm}$$

$$Q = m \cdot L_v$$

Latent heat of fusion (L_f) ⇒

Amt of heat req. to change 1gm ice into 1gm H₂O at its M.P.

$$(L_f)_{\text{ice}} = 80 \text{ cal/gm}$$

$$Q = m \cdot L_f$$

Important Data ⇒

$$s_{\text{ice}} = s_{\text{steam}} = 0.5 \text{ cal/gm}^{\circ}\text{C}$$

$$s_{\text{water}} = 1 \text{ cal/gm}^{\circ}\text{C}$$

Eg: 10kg ice @ -10°C mixed with 40kg water @ 45°C.

$$\begin{aligned} Q_{\text{ice}} &= m s \Delta T = 10 \times 0.5 \times 2100 \times 10 = 21 \times 10^4 \text{ J} \\ Q_{\text{water}} &= m \cdot L_f \Rightarrow 10 \times 80 \times 10^3 \times 4.2 = 336 \times 10^4 \text{ J} \end{aligned}$$

Now,

$$Q = m s \Delta T \Rightarrow 40 \times 4200 \times 45 = 756 \times 10^4 \text{ J}$$

$$Q_{\text{absorb}} = 351 \times 10^4 \text{ J} \quad Q_{\text{release}} = 756 \times 10^4 \text{ J} \quad \Delta Q = 399 \times 10^4 \text{ J}$$

$$Q = m s \Delta T \Rightarrow 399 \times 10^4 = 50 \times 4200 \times \Delta T \Rightarrow \Delta T = 19^{\circ}\text{C}$$

Heating Curve :

$$\Delta T = \frac{Q}{m \cdot S}$$

$$\Rightarrow T_f = \frac{1}{m \cdot S} (Q) + T_i \quad \left[\text{slope} = \frac{1}{m \cdot S} \right]$$

